Crew Utilization, FTL & Crew Ratio Assessment

(Including Wheel Ops Separation, Network Growth & Aircraft Utilization)

1. Crew Ratio & Sub-Fleet Position (Clarified)

We are **not effectively managing two sub-fleets** with the current Captain and First Officer strength. Although there is a perception of a Gan / MLE split, operationally the crew pool continues to function as a **single constrained system**, governed by identical regulatory limitations and commercial demand.

A further constraint exists due to the **release of 2 Captains and 1 First Officer to Wheel Operations**. These crews are based separately and are **not interchangeable with Float Operations**, effectively reducing available Float Ops crew strength without any corresponding reduction in flying demand.

2. Current Operational Reality

Since **October 2025**, commercial demand has required **9–10 aircraft to be active daily** (average **9.25 aircraft/day**), including Gan operations.

Crew utilization levels are already high:

- **~95% of Flight Time Limit (FTL)**
- **~90%+ of Flight Duty Period (FDP)**

At these levels, there is **no meaningful operational buffer**, regardless of aircraft base or assignment.

3. Leave & Availability Constraints

- **3 Captains are currently on leave (1 out of leave and working on our request, bringing down captains on leave to 2)**
- Operational planning requires **5 Captains to be on leave daily**
- Leave is being **delayed** to protect schedule integrity

Despite delaying leave:

- FTL and FDP limitations remain unresolved
- Individual Captains continue to reach regulatory limits
- This approach is **unsustainable and cannot be relied upon long-term**

4. Aircraft Split Does Not Reduce Crew Requirement

Only **one aircraft** is assigned to Gan from a **10-aircraft fleet**. Wheel Ops and MLE Ops are therefore **not independent sub-fleets**, and the same regulatory framework applies across the operation:

- FTL
- FDP
- Sector limits
- SPI / SPT fatigue thresholds

Conclusion: The crew ratio does **not change** due to base or aircraft split, as crew utilization is already maximized under combined operational demand.

5. Network Growth Impact

The operation continues to expand without a corresponding increase in crew capacity.

- **Four new destinations** have already been added across 2025–2026
- Additional destinations
  - Hideaway
  - Fulhadhoo
  - Kamadhoo

  - Velidhoo
  - Thoddoo

This growth directly increases:

- Flight Time Limitations
- Sector count
- FDP accumulation
- Fatigue exposure

6. Block Hours & Aircraft Utilization Evidence

Analysis of actual utilization (Nov 2024 – Jan 2026) and forward commercial forecasts shows:

- **Average block hours / month:** ~1,512 hrs
- **Average block hours / day:** ~50.4 hrs
- **Aircraft utilized per day:** ~9.25
- **FDP allocated per aircraft:** ~13.5 hrs

Flying demand is **consistent and sustained**, not seasonal, with FDP exposure trending toward **13:30 per aircraft**, leaving minimal margin.

7. Crew Ratio Justification – Simple Calculation

To validate whether the **2.4-2.5 crew ratio** remains appropriate, a simple calculation based on **actual utilization data** is applied below.

Daily Flying Demand

Item	Value
Avg aircraft utilized per day	~9.25 (≈10)
Avg block hours per day	~50.4 hrs
FDP allocated per aircraft	~13.5 hrs
Total FDP demand per day	**~135 hrs**

Crew Required – FTL Constraint

Parameter	Value	
FTL limit	100 hrs / 28 days	
Allowable avg hrs per crew	~4.5 hrs/day	Maintain 28 Days Limit and SPT @90%
Crew required (FTL)	50.4 ÷ 4.5 ≈ 11.2 crews/day	

Crew Required – FDP Constraint *(Limiting Factor)*

Parameter	Value	
Total FDP demand per day	~135 hrs	
Max sustainable FDP per crew	~10.5 hrs/day	Maintain 7 Days/ 14Days/ 28 Days restriction with Exceeding SPT
Crew required (FDP)	**135 ÷ 10.5 ≈ 12.85 crews/day**	

Average Daily Flying Crew Requirement

Calculation	Result	
FTL-based crews	~11.2	Required to Maintain- FDP, FTL, SPI, Sector, Standby Crew (Cover Absenteeism)
FDP-based crews	~12.85	
Average flying crews required	**~12 crews/day**	

Required Rest Period and Duty other than Normal Flying

Item	Crew / Day	
Leave	5.0	22 Crew, 75 Days, 90 Days Leave Pattern / 365 Days
Weekly days off	5.0	

		22 Crew = Rostered Week * 2 Day off (Recurrent Recovery Rest) / 365 Days
Admin duties (CP / Training Captains)	~1.3	3 Days Admin (24hrs of Admin hours) per Week
Flight Training & checks / Ground Training / CBT / Instructor	~1.3	Assigned all Checks / Rides/ Flight Training (Excluding Ground and CBT) / Number of Eligible Working Days
Total unavailable crew	**~12.6 crews/day**	`

Total Crew Requirement & Ratio Outcome

Item	Value
Flying crews required daily	~12.0
Non-flying crews daily	~12.6
Total crews required	**~24.6 crews**
Aircraft in operation	9.25–10
Crew ratio	**≈ 2.4 - 2.5 per aircraft**

8. Other Factors

The 2.4-2.5 ratio already accounts for:

- Leave and Recurrent Recovery Rest Period (Day off)
- Training and checking
- Administrative roles

Any further network or sector growth will **increase** the required ratio, not reduce it.

9. Overall Conclusion

- The operation is running at **maximum sustainable utilization**
- Releasing **2 Captains and 1 F/O to Wheel Ops** reduces Float Ops flexibility
- Aircraft utilization, flights per day, and sectors per day continue to increase
- The **2.5 crew ratio remains the minimum requirement**
- Operating below this ratio leads to:
  - Leave deferrals
  - FTL/FDP exceedances
  - Predictable fatigue and schedule risk
  - Risking FRMS commitments

10. Recommendation

To stabilise utilization, restore resilience, and protect regulatory compliance:

Recommended Action

- Hire:
  - **2 Captains**
  - **2 First Officers**
- Maintain **minimum 2.5 crew ratio** for DHC-6 Operations
- Treat Wheel Ops crews as **non-available** for Float Ops planning

If 4 caps are maxed out. To asses if 2 new hires are sufficient, I think we should also include the rolling 28-day flight time trend. If the majority of the captains are trending towards their 900 hour annual limit, the 2 external hires will only provide temporary reprieve. With the trend (which we already have I believe though our system), can we project the FTL cliff in the coming months?

Report Attached of FTL

Yes, the rolling **28-day flight time trend** should be included to assess whether **2 new Captain hires** are sufficient. However, it is important to note that **flight time is not the only limiting factor** in the current operation.

While **4 Captains are already at or near FTL limits**, the more restrictive constraint is **Flight Duty Period (FDP)**, which is governed by **multiple overlapping monitoring limits** (7-day, 14-day, and 28-day FDP), in addition to sector caps and SPI/SPT fatigue thresholds.

The operation is currently sustaining **9–10 aircraft daily utilization**, with increasing sectors per day and extended duty days. This results in **FDP accumulation occurring faster than flight time accumulation**, particularly under high sector density operations. As a result, crews are often constrained by **FDP limits before reaching flight time limits**.

The rolling trend shows that the **majority of Captains are trending upward** in both FDP and flight time exposure. In this context, the addition of **2 external Captains does not reduce utilization**; it is required to **restore the minimum crew level** needed to operate the existing schedule. It does not unwind accumulate FDP or flight time, nor does it create operational buffer.

Using the rolling FDP and flight time data already available in the system, we can project:

- The **point at which FDP limits become binding** for additional Captains
- The **progression toward 28-day and annual FTL limits**
- The resulting **FTL/FDP cliff**, where crew availability reduces faster than crew additions can offset

I think we should support with the duty and flight time report for the last quarter and the projections for this quarter inclusive of the cadets training and the command training with current scenario. Then we should propose our best case scenarios mapped on to this. This will give is the so-called 2.5 ratio still holds with this divided fleet operations in Gan and MLE. I think with this change we should do away with the 2.5 ratio as it won't make sense.

To support this attached the **last quarter duty and flight time reports**, together with **current-quarter projections**, explicitly including **cadet training and command training** under the present operating model. However, when this data is applied to the actual utilization figures we already have, it does **not support doing away with the 2.4-2.5 crew ratio**.

Based on the data reviewed:

- **Aircraft utilization averages 9.25 aircraft per day**, trending to **10 aircraft** since Oct-25
- **Average block hours are ~50.4 hrs per day**, with **~1,512 hrs per month**
- **FDP allocated per aircraft is ~13.5 hrs**, resulting in **~125–135 FDP hours per day**
- **Sectors per day have increased materially**, from the mid-80s to **115–119 sectors/day**
- **Flights per day increased from ~52 (2024) to ~64 (2026)**, a **23% increase**
- **4 Captains are already maxed**, despite leave deferrals and delayed training

When cadet line training and command training are included:

- Training Captains are fully removed from training to make availability to line
- FDP and flight time are concentrated on fewer line Captains
- FDP limits (monitored across **7-day, 14-day, and 28-day windows**) become binding **before** flight time in many cases

In this context, dividing operations between **Gan and MLE** does not create efficiency. The data shows the opposite:

- The fleet is still being operated at **near-maximum utilization**
- The same FDP, FTL, sector, and SPI/SPT limits apply system-wide
- Releasing **2 Captains and 1 F/O to Wheel Ops** further reduces usable Float Ops crew without reducing demand

When the actual figures are applied:

- **~12 crews/day** are required purely to cover flying (FTL/FDP driven)
- **~12 crews/day** are unavailable due to leave, off days, admin, and training
- This yields a requirement of **~24 crews for ~9.25–10 aircraft**
- Resulting in a ratio of **~2.4 -2.5 crew per aircraft**

Conclusion

Including last-quarter actuals and current-quarter projections with cadet and command training will strengthen the analysis. However, based on the utilization, FDP exposure, sector growth, and training impact already evident in the data, the **2.4- 2.5 crew ratio remains a minimum operational requirement**. Doing away with it under a divided Gan/MLE footprint would **understate crew demand** and increase FDP, FTL, and fatigue risk rather than reflect operational reality

Block Hours Distribution report 2024-2026

Months	Total monthly	Daily	% Increase or Decrease		Captain Availability
Oct-24	1128:25:00	40:18:02			
Nov-24	1374:40:00	49:05:43	8:47:41	18%	23
Dec-24	1538:05:00	54:55:54	14:37:51	27%	23
Jan-25	1656:20:00	59:09:17	18:51:15	32%	23
Feb-25	1555:05:00	55:32:19	15:14:17	27%	23
Mar-25	1452:25:00	51:52:19	11:34:17	22%	23

Apr-25	1501:00:00	53:36:26	13:18:24	25%	23
May-25	1214:55:00	43:23:24	3:05:21	7%	23
Jun-25	1141:35:00	40:46:15	0:28:13	1%	23
Jul-25	1279:15:00	45:41:15	5:23:13	12%	23
Aug-25	1396:25:00	49:52:19	9:34:17	19%	23
Sep-25	1179:25:00	42:07:19	1:49:17	4%	23
Oct-25	1477:50:00	52:46:47	12:28:45	24%	23
Nov-25	1534:00:00	54:47:09	14:29:06	26%	22
Dec-25	1615:30:00	57:41:47	17:23:45	30%	22
Jan-26	1694:25:00	60:30:54	20:12:51	33%	22

Sectors of Variation. 2024- 2026

Month	Sum of Leg	Sector / Day	+/-	% Increase / Decrease
Oct-24	2527.00	84.23		
Nov-24	2958.00	98.60	14.37	15%
Dec-24	3216.00	107.20	22.97	21%
Jan-25	3318.00	110.60	26.37	24%
Feb-25	3216.00	107.20	22.97	21%
Mar-25	3156.00	105.20	20.97	20%
Apr-25	3190.00	106.33	22.10	21%
May-25	2598.00	86.60	2.37	3%
Jun-25	2414.00	80.47	-3.77	-5%
Jul-25	2599.00	86.63	2.40	3%
Aug-25	2726.00	90.87	6.63	7%
Sep-25	2468.00	82.27	-1.97	-2%
Oct-25	3088.00	102.93	18.70	18%
Nov-25	3329.00	110.97	26.73	24%
Dec-25	3446.00	114.87	30.63	27%
Jan-26	3572.00	119.07	34.83	29%

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